MIL-M-28915(MC)
21 March 1969

MILITARY SPECIFICATION

MODEM, TELEGRAPH-TELEPHONE: MD-791/GCC

1. SCOPE

1.1 This specification covers requirements for one type telegraph-telephone modem.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-P-15024	-	Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555	-	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-21981	-	Electronics Type Designations, Identification Plates and Markings; Requirements for.
MIL-R-22732	-	Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.

FSC 5805

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MIL-STD-188 - Military Communication System Technical Standards.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-826 - Electromagnetic Interference Test Requirements and Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The Modem, Telegraph-Telephone, MD-791/GCC, shall consist of a device capable of combining two or three Terminal, Telegraph-Telephones, TH-81/GCC, so that they operate simultaneously over one radio path. Definitions for abbreviations not defined herein may be obtained from MIL-STD-188.

3.1.1 Type of function of equipment. The modem shall be a lightweight, solid state high performance equipment operating in 300-hertz (Hz) to 64-kilohertz (kHz) frequency range. The equipment shall also provide the following:

(a) Compatibility with Terminal, Telegraph-Telephones, TH-81/GCC, Receiver-Transmitter, RT-524/VRC, and Receiver, R-442/VRC.

(b) Low power consumption.

(c) High reliability and ease of maintainability.

(d) Small physical size.

3.2 First article. Prior to submission of any complete production units for acceptance, the contractor shall furnish first articles for testing under the direction and supervision of the cognizant Government inspector to determine conformance to this specification. First article is defined as a production sample which will be identical to the production items which the contractor will subsequently produce in fulfillment of the contract. Unless otherwise specified, a quantity of five first articles shall be furnished (see 6.2).

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Approval of the first articles by the activity concerned shall not relieve the contractor of the responsibility to supply equipment conforming to this specification. Examination and testing shall be as specified herein. Any changes or deviations of the production units from the first articles shall be subject to the approval of the contracting officer. The items subjected to first article tests may be included in the final shipment as part of the total quantity to be supplied under the contract, provided they have been completely refurbished.

3.3 Reliability assurance. Reliability shall receive major emphasis during the design, fabrication and construction of units. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The first article, production and sample items shall be subjected to the mechanical and electrical tests specified in Section 4. Mechanical wear shall be such that mechanical or functional performance is not affected. Abrasion of painted surfaces is allowable. Electrical performance of the unit shall not be degraded. The contractor is required to demonstrate that the terminal is capable of calculated mean time between failure (MTBF) of not less than 15,000 hours. The contractor is required to demonstrate actual MTBF of not less than 500 hours under standard and service conditions. Test level No. 2 of MIL-R-22732 shall be utilized in establishing demonstrated test conditions.

3.4 Design and construction. Design and construction shall be as follows.

3.4.1 Mechanical parts. Where mechanical parts are employed, whose functions are to rotate, cam or oscillate against other mechanical parts and the best choice of materials includes other than those specified in MIL-E-16400, use of the other materials may take precedence.

3.4.2 Parts and materials. In the selection of parts and materials, the fulfillment of major objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus-resistance and nontoxicity as specified in MIL-T-152 shall be met.

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3.4.3 Nonstandard parts. When it has been determined that requirements cannot be met by use of parts and materials specified in MIL-E-16400, the contractor shall immediately inform the contracting officer by submission of detailed data as required by MIL-E-16400, and shall request approval.

3.4.4 Components. Parts and materials approved by specifications listed in MIL-E-16400 shall be given first consideration.

3.4.5 Case material. The construction of the case and front panels for the unit shall be made of cast aluminum alloy. Where panels are removable from the basic case and interconnecting cables permanently enter the case, the openings shall be protected with gaskets and packing to assure watertight protection for the equipment in transit. The controls and cable connectors shall be splashproof in operation.

3.4.6 Treatment and painting. Treatment and painting shall conform to MIL-T-704. Color shall match Marine Corps green, No. 23 (see 6.3).

3.4.7 Standard conditions. The following conditions shall be used as the basis for establishing normal requirements for conducting laboratory and bench test of the unit.

(a) Room ambient temperature range +20° Centigrade (C) to +40° C.
(b) Altitude, normal ground.
(c) Room ambient humidity up to 95 percent relative.
(d) Input voltage, 24 direct current volts (Vdc) ±5 percent.

3.4.8 Service conditions. The unit shall operate satisfactorily under any of the following service conditions:

(a) Ambient temperature range -40° C to +65° C.

(b) Altitude, sea level to 10,000 feet.

(c) Ambient humidity range 0 to 95 percent relative.

(d) Vibration (see 4.3).

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(e) Shock (see 4.4).

(f) Input voltage 22 to 32 Vdc.

(g) Immersion (leakage) (see 4.5).

(h) Salt fog environment (see 4.11).

(i) Blowing sand and dust (see 4.12).

(j) Temperature shock (4.8).

(k) High temperature (see 4.6).

(l) Low temperature (see 4.7).

3.4.9 Storage conditions. The equipment shall be capable of being stored under the following conditions with a subsequent period of operation:

(a) Ambient temperature range -62° C to +71° C
(b) Altitude - Sea level to 50,000 feet

3.4.10 Stability.

3.4.10.1 Electrical. The equipment shall operate as specified herein for at least 500 hours, either continuously or intermittently, independent of component failure without necessity for readjustment of any controls other than those located on the front panel.

3.4.10.2 Mechanical. All controls and adjustments shall be designed so that their position of adjustment will not be altered when the equipment is subjected to the specified service conditions.

3.4.11 Interchangeability. All components and parts shall be physically, functionally, mechanically and electrically interchangeable with any other unit produced under this specification.

3.4.12 Identification marking. Subassemblies shall be identified with the manufacturer's part numbers, steel stamped, rubber stamped or silk screened on a fixed surface

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if available, or on a permanent tag affixed in a manner which shall not impair operation of the assembly, and shall not be easily removable. This marking shall be visible when mounted in the assembly.

3.5 Fungus resistance. The unit shall be rendered fungus proof by the selection of parts and materials which are nonnutrient for fungus.

3.6 Electrical characteristics. The electrical characteristics shall be as specified in the following paragraphs for a nominal input voltage of 24 Vdc.

3.6.1 Modem capability. The modem shall be capable of being used with two or three Terminal, Telegraph-Telephones, TH-81/GCC.

3.6.2 Transmit.

3.6.2.1 Input signal level. The transmit input signal for each group shall be 0 dBm ±1 decibel (dB) and the pilot shall be -10 dBm ±1 dB from the Terminal, Telegraph-Telephone TH-81/GCC flat output.

3.6.2.2 Input impedance. The input impedance of each group shall be 600 ohms ±10 percent over the input frequency range of 300 Hz to 19.7 kHz.

3.6.2.3 Input frequency range. The transmit input frequency range shall be 300 Hz to 19.7 kHz for each of the three groups.

3.6.2.4 Output signal level. The group 1 line level shall be -10 dB ±1.5 dB, group II -7 dB ±1.5 dB and Group III -4 dB ±1.5 dB when loaded with 600-ohm ±1 percent impedance.

3.6.2.5 Transmit output impedance. The transmit output impedance shall be 600 ohm ±10 percent over the output frequency range 300 Hz to 64 kHz.

3.6.2.6 Transmit output frequency range. The transmit frequency output range shall be 300 Hz to 64 kHz.

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3.6.2.7 Carrier leak. The group carrier leak shall be -20 dBm maximum.

3.6.3 Receive.

3.6.3.1 Receive input signal level. The group I receive input signal level shall be -10 dB ±1.5 dB, group II -7 dB ±1.5 dB and group III -4 dB ±1.5 dB.

3.6.3.2 Receive input impedance. The group receive input impedance shall be 600 ohms ±10 percent.

3.6.3.3 Receive input frequency range. The input frequency range shall be 300 Hz to 64 Hz.

3.6.3.4 Receive output signal level. The three sub-group receive output signal levels shall be -20 dB ±5 dB.

3.6.3.5 Receive output impedance. The three sub-group receive output impedances shall be 300 ohms maximum over the frequency range 300 Hz to 19.7 kHz.

3.6.3.6 Receive output frequency range. The output frequency range for each group shall be 300 Hz to 19.7 kHz.

3.6.3.7 Receive carrier leak. The sub-group carrier leak shall be -20 dB maximum below the actual test tone level specified in 3.6.3.4.

3.6.4 Distortion.

3.6.4.1 Intermodulation distortion. The intermodulation distortion shall be 50 dB below test tone.

3.6.4.2 Cross modulation. The cross modulation shall be 50 dB below test tone.

3.6.5 Frequency translation. The group I translation characteristics shall be lower side band, 0.3 kHz to 19.7 kHz bandwidth, the group II 20.3 kHz to 39.7 kHz and the group III 44.3 kHz to 63.7 kHz.

3.6.6 Synchronization. The modulator and demodulator local oscillators frequencies shall be 40 kHz for group II and 64 kHz for group III and shall be synchronized to the group I pilot frequency.

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3.6.7 Input power. The input power consumption of the Modem, MD-791/GCC, shall not exceed 1.5 watts not including the Terminal, Telegraph-Telephones, TH-81/GCC.

3.7 Connectors. The unit shall have four captive cable assemblies for connection to the three Terminal, Telegraph-Telephones, TH-81/GCC, and the associated radio set.

3.8 Dimensions. The maximum external dimensions of the modem shall not exceed 4.593 inches wide by 9.25 inches deep by 2.5 inches high.

3.9 Weight. The weight shall not exceed 7-1/2 pounds.

3.10 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where possible. Drawings for identification plates shall be submitted for approval 90 days prior to first article tests. Nomenclature shall be requested in accordance with MIL-E-21981 and shall be submitted 90 days prior to the first article testing.

3.11 Workmanship. Units covered by this specification shall be manufactured and finished in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection and tests. Inspection and tests shall consist of all tests specified herein to demonstrate that the units submitted for approval meet all requirements of this specification. First article testing shall be witnessed by a representative of the contracting officer.

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4.2.1 Tests. Table I lists the tests required for first article, production and sample tests.

4.2.1.1 Inspection and test procedures. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for approval 30 days prior to the initiation of any official testing.

4.2.1.2 Tests to be performed. Units to be tested shall be the quantities indicated.

(a) First article. Unless otherwise specified, a quantity of five complete units shall be used to complete these tests (see 3.2). Tests performed on one unit need not be repeated on another. The Government representative will select which unit will be subjected to each test.

(b) Production. All units shall be subjected to the production tests.

(c) Samples. Sample units shall be randomly selected by the cognizant Government inspector.

Qty. produced	Qty. to be tested
1-49	1
50-99	2
100-149	3
150-199	3
200-300	5
	One additional unit per 100 units.

4.3 Vibration test. One unit shall be subjected to the following vibration test with the unit operating. The unit shall be affixed to the vibration table and be subjected to the following conditions. Vibration in the frequency range of 10 to 55 Hz at 5 gravity (g) or 0.060 inch total excursion, whichever is the limiting value. The frequency change shall be at a rate of less than 1 Hz. The duration of the test shall be 1/2 hour in each of three planes. There shall be no rigid body resonances and no signs of internal or external physical deterioration.

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Table I Tests

Electrical Para	Environmental Para.	
Transmit 3.6.2	Standard condition	3.4.7
Receive 3.6.3	Vibration	4.3
Intermodulation distortion 3.6.4.1	Shock	4.4
Cross modulation 3.6.4.2	Immersion	4.5
Power Consumption 3.6.7	High temp.	4.6
Interference 4.13	Low temp.	4.7
	Humidity	4.9
	Low pressure	4.10
	Salt fog	4.11
	Sand and dust	4.12
	Temp. shock	4.8

Transmit	Receive	Intermodulation distortion	Cross modulation	Power Consumption	Interference
FA P	FA P	FA	FA	FA P	FA
FA	FA				
FA	FA				
FA S	FA S	FA S	FA S	FA S	
FA S	FA S	FA S	FA S	FA S	
FA	FA				
FA	FA				
FA	FA				
FA	FA				
FA	FA				

NOTE: FA= First article S= Sample test P= Production test

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4.4 Shock test. One unit shall be subjected to the following shock test with the unit operating. The unit shall be mounted rigidly on the adapter plate of the LAB package testing machine. The machine shafts shall be operated in phase and adjusted in speed to produce average peak accelerations of 10 g's with occasional peaks of 20 g's as measured on the case of the equipment. Malfunction is permissible provided that normal operation can be restored by adjustment of front panel controls. There shall be no signs of internal or external physical deterioration.

4.5 Immersion (leakage) test. The unit shall be submerged in water to a depth of 3 feet, measured from its highest point. The unit shall be held in this position for 1 hour, and during the test all panel controls shall be operated three times through their full travel. There shall be no entry of water into the unit. There shall be no signs of internal or external physical deterioration.

4.6 High temperature test. The unit shall be tested in accordance with procedure I, method 501, of MIL-STD-810. The unit shall be operated at a temperature not less than 60° C nor greater than 65° C.

4.7 Low temperature test. The unit shall be tested in accordance with procedure I, method 502, of MIL-STD-810. The unit shall be operated at a temperature not greater than -40° C.

4.8 Temperature shock test. The unit shall be tested in accordance with procedure I, method 503, of MIL-STD-810 with exception that the maximum temperature shall be 71° C.

4.9 Humidity test. The unit shall be tested in accordance with procedure I, method 507, of MIL-STD-810.

4.10 Low pressure test. The unit shall be tested in accordance with procedure I, method 500, of MIL-STD-810.

4.11 Salt fog test. The unit shall be tested in accordance with procedure I, method 509, of MIL-STD-810. Salt deposits shall be removed by rinsing with tap water.

4.12 Sand and dust test. The unit shall be tested in accordance with procedure I, method 510, of MIL-STD-810.

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4.13 Interference test. The equipment shall be tested for conducted and radiated interference in accordance with MIL-STD-826.

4.14 Test reports. The contractor shall prepare and submit to the procuring activity copies of all test reports not less than 10 days after completion of tests.

4.15 Approval. Approval of the first articles shall be by the contracting officer upon satisfactory completion of all tests.

4.16 Acceptance. With exception of the first articles, acceptance of all items shall be by the cognizant Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A or C in accordance with MIL-E-17555 as specified in the contract or order (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This unit is to be employed with two or three Terminal, Telegraph-Telephones, TH-81/GCC, to expand the number of telemetered channels under tactical conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of first articles required (see 3.2 and 4.2.1.2(a)).
(c) Level of preservation, packaging, and packing (see 5.1).

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6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps, (Code CSY-10), Headquarters, U. S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first articles shall be examined and tested for approval, at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the contracting officer.

Preparing activity:
Navy - MC
Project No. 5805-N135

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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